30 Semiconductors

Boolean Algebra

1. If $A = 1$, $B = 0$ then in terms of Boolean algebra $A + \overline{B}$ equal to:

   (*a*) A (*b*) B

   (*c*) $\overline{A}$ (*d*) $\overline{A + B}$

2. Which of the following is not equal to 1 in Boolean algebra:

   (*a*) A + 1 (*b*) $A + \overline{A}$

   (*c*) $A \,.\, \overline{A}$ (*d*) $\overline{A \,.\, \overline{A}}$

3. What will be the input of A and B for the Boolean expression $(\overline{A + B}).(\overline{A \,.\, B}) = 1$?

   (*a*) 0, 0 (*b*) 0, 1

   (*c*) 1, 0 (*d*) 1, 1

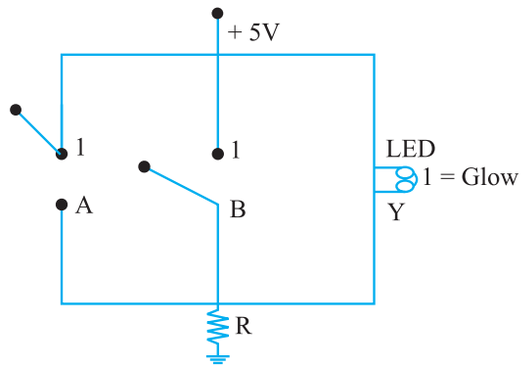
4. Which of the following Boolean expression is not correct:

   (*a*) $\overline{\overline{A}.\overline{B}} = A + B$ (*b*) $\overline{\overline{A} + \overline{B}} = A \,.\, B$

   (*c*) $\overline{\overline{A \,.\, B}} = A \,.\, B$ (*d*) $\overline{1} + \overline{1} = 1$

Logic Gate

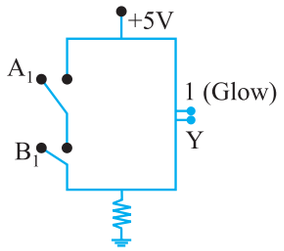
5. Name the logic gate equivalent to the diagram attached.



   (*a*) OR (*b*) NOR

   (*c*) NAND (*d*) AND

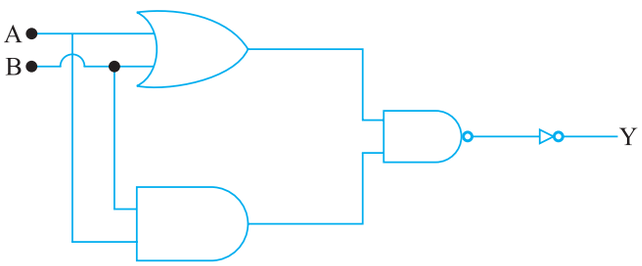
6. The logic gate equivalent to the given circuit diagram is:



   (*a*) OR (*b*) NAND

   (*c*) NOR (*d*) AND

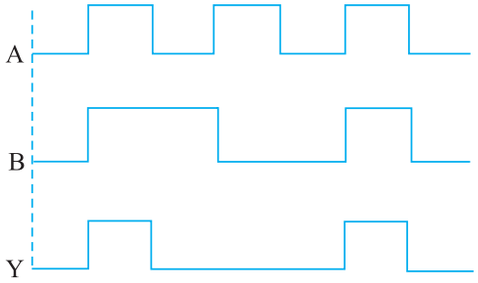
7. The logic operations performed by the given digital circuit is equivalent to:



   (*a*) AND (*b*) NOR

   (*c*) OR (*d*) AND

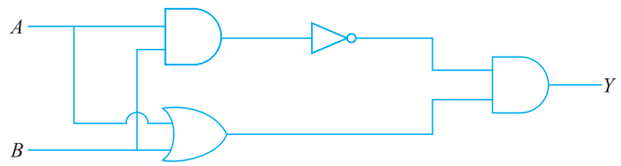
8. A logic gate circuit has two inputs A and B and output Y. The voltage waveforms of A, B and Y are shown below. The logic gate circuit is:



   (*a*) AND gate (*b*) OR gatae

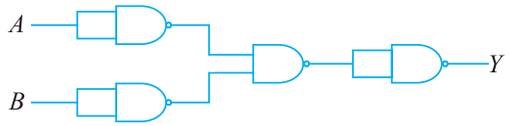
   (*c*) NOR gate (*d*) NAND gate

9. Which one of the following will be the output of the given circuit?



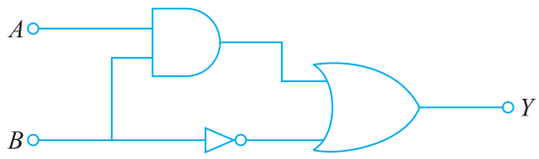
(*a*) XOR gate (*b*) NOR gate
(*c*) NAND gate (*d*) AND gate

10. The following logic gate is equivalent to:



(*a*) NOR gate (*b*) NAND gate
(*c*) OR gate (*d*) AND gate

11. Find the truth table for the function Y of A and B represented in the following figure:



(*a*)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

(*b*)

A	B	Y
0	0	1
0	1	0
0	1	0
1	1	1

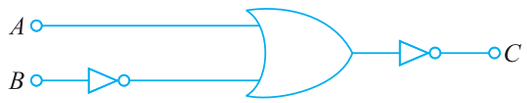
(*c*)

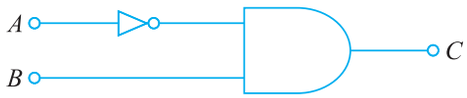
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

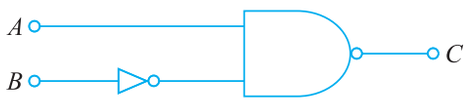
(*d*)

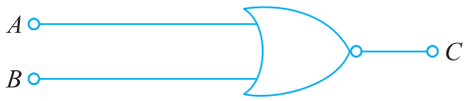
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

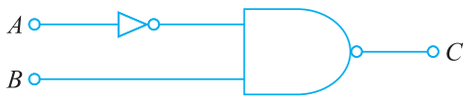
12. The logic circuit shown above is equivalent to:



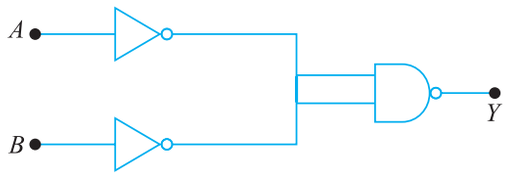
(*a*)


(*b*)


(*c*)


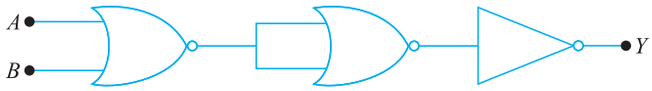
(*d*)


13. The logic gate equivalent to the given logic circuit is:



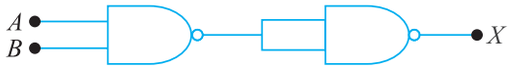
(*a*) OR (*b*) AND (*c*) NOR (*d*) NAND

14. The given electrical network is equivalent to



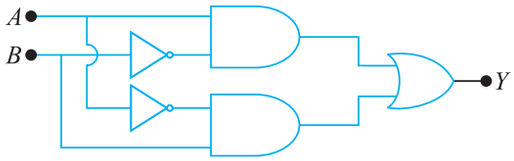
(*a*) OR gate (*b*) NOR gate(*c*) NOT gate(*d*) AND gate

15. The output (X) of the logic circuit shown in figure will be



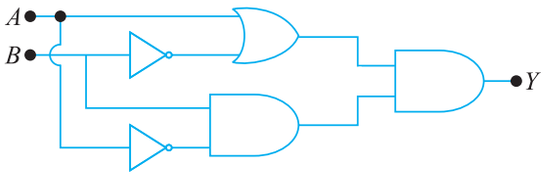
(*a*) $X = A.B$ (*b*) $X = \overline{A + B}$
(*c*) $X = \overline{\overline{A}.\overline{B}}$ (*d*) $X = \overline{A.B}$

16. In the combination of the following gates the output Y can be written in terms of inputs A and B as



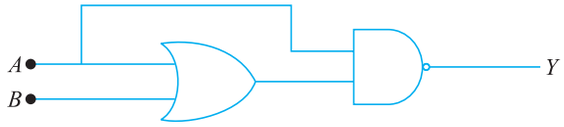
(*a*) $\overline{A.B}$ (*b*) $A.\overline{B} + \overline{A}.B$
(*c*) $\overline{A.B} + A.B$ (*d*) $\overline{A + B}$

17. The output Y of the following circuit for given Input is



(*a*) $A.B(A + B)$ (*b*) $A.B$
(*c*) 0 (*d*) $\overline{A}.B$

18. The truth table for the circuit given in the figure is:



(*a*)

A	*B*	*Y*
0	0	1
0	1	1
1	0	0
1	1	0

(*b*)

A	*B*	*Y*
0	0	0
0	1	0
1	0	1
1	1	1

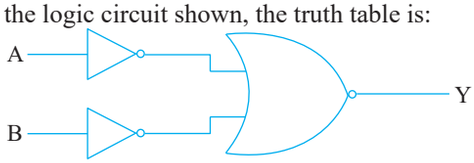
(c)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(d)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

19. For the logic circuit shown, the truth table is:



(a)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(b)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

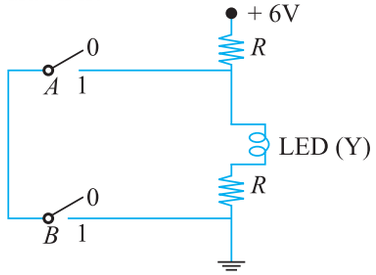
(c)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(d)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

20. The correct Boolean operation represented by the circuit diagram drawn is:



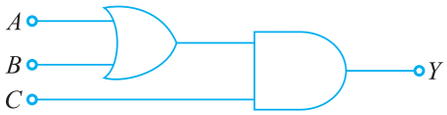
(a) NOR (b) AND (c) OR (d) NAND

21. What is the output Y in the following circuit, when all the three inputs A, B, C are first 0 and then 1?



(a) 0, 1 (b) 0, 0 (c) 1, 0 (d) 1, 1

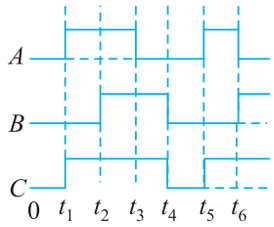
22. To get output 1 for the following circuit, the correct choice for the input is



(a) $A = 1, B = 1, C = 0$ (b) $A = 1, B = 0, C = 1$

(c) $A = 0, B = 1, C = 0$ (d) $A = 1, B = 0, C = 0$

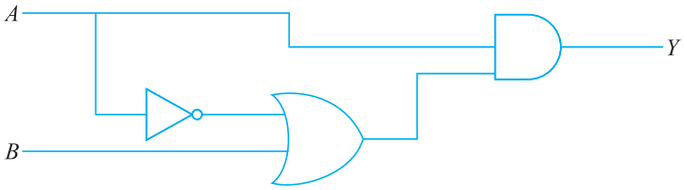
23. The figure shows a logic circuit with two inputs A and B and the output C. The voltage wave forms across A, B and C are as given. The logic circuit gate is



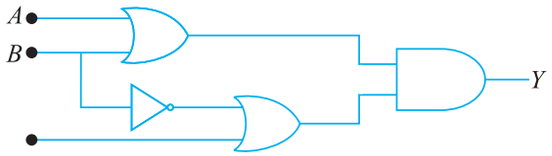
(a) OR gate (b) NOR gate

(c) AND gate (d) NAND gate

24. Name of gate:



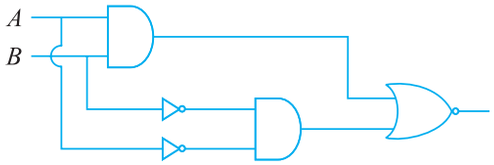
25. Find out value of Y



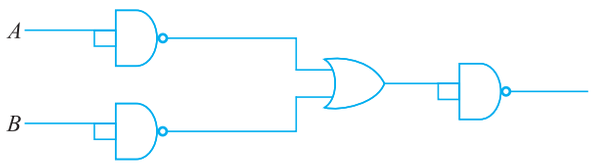
(a) $(\overline{A}+B).(\overline{B}+C)$ (b) $(A+B).(\overline{B}+C)$

(c) $(A+\overline{B}).(B+\overline{C})$ (d) $(\overline{A}+B).(\overline{B}+\overline{C})$

26. What is the name of gate:

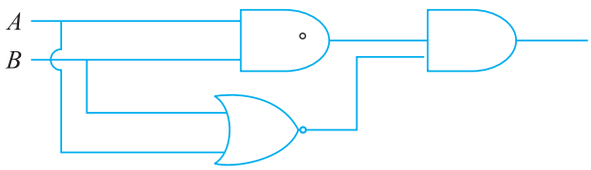


27. What is the name of gate:

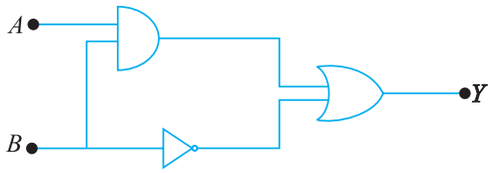


(a) OR (b) NOR (c) NAND (d) AND

28. What is the name of gate:



29. In the figure shown if $A = 1$ and $B = 0$ then y will be



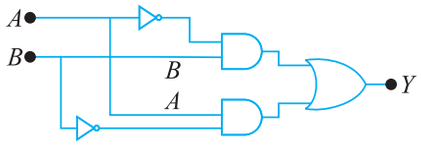
(*a*) 0

(*b*) 1

(*c*) 2

(*d*) Any of these

30. The truth table for the following logic circuit is



(*a*)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(*b*)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

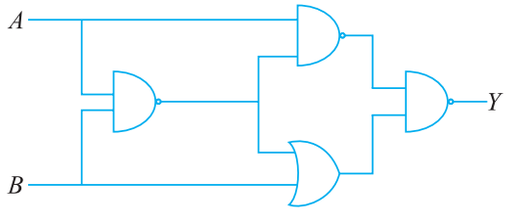
(*c*)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(*d*)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

31. The output of the given logic circuit is:



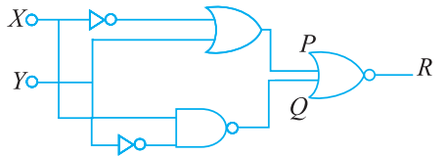
(*a*) $A\bar{B} + \bar{A}B$

(*b*) $AB + \overline{AB}$

(*c*) $A\bar{B}$

(*d*) $\bar{A}B$

32. The get output 1 at R, for the given logic gate circuit the input values must be:



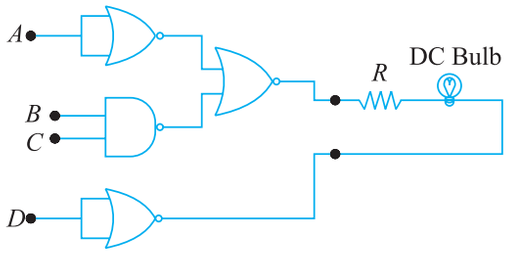
(*a*) $X = 0, Y = 1$

(*b*) $X = 1, Y = 1$

(*c*) $X = 1, Y = 0$

(*d*) $X = 0, Y = 0$

33. Following gates section is connected in a complete suitable circuit.



For which of the following combination, bulb will glow (ON):]

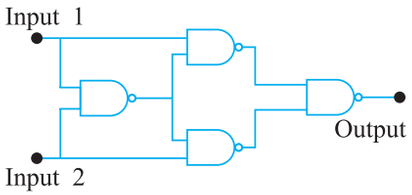
(*a*) $A = 0, B = 1, C = 1, D = 1$

(*b*) $A = 1, B = 0, C = 0, D = 0$

(*c*) $A = 0, B = 0, C = 0, D =$

(*d*) $A = 1, B = 1, C = 1, D = 0$

34. For the following combination of gates select the correct statement



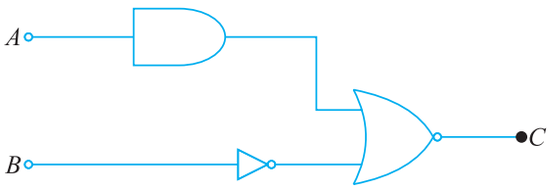
(*a*) The output is 1 when both the inputs are 1

(*b*) The output is 1 when both the inputs are 0

(*c*) The output is 0 when the two inputs are differ

(*d*) The output is 1 when the two inputs are different

35. Find the truth table for the function Y of A and B represented in the following figure,



(*a*)

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

(*b*)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	1

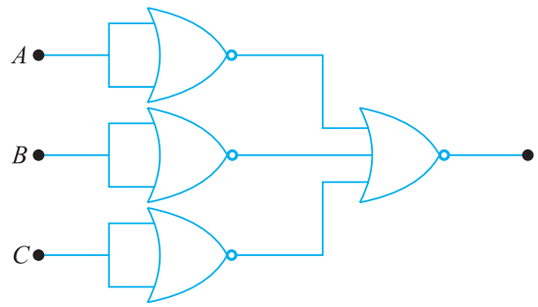
(*c*)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

(*d*)

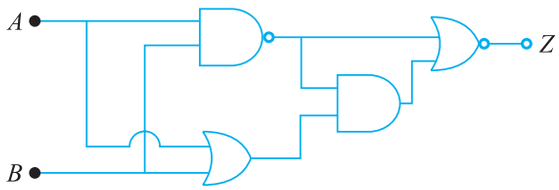
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

36. Identify the operation performed by the circuit given below



(*a*) AND (*b*) NOT (*c*) NAND (*d*) OR

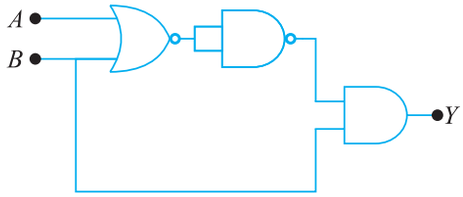
37. In the following, digital circuit, what will be the output at '*Z*', when the input (*A*, *B*) are (1, 0), (0, 0), (1, 1), (0, 1)



(*a*) 0, 1, 0, 0 (*b*) 1, 0, 1, 1

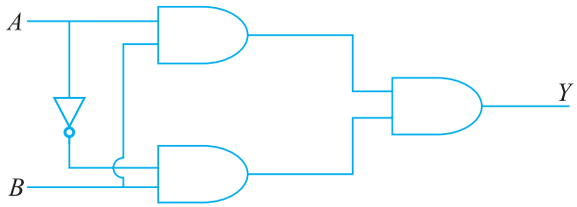
(*c*) 1, 1, 0, 1 (*d*) 0, 0, 1, 0

38. The output *Y* of the given network is



(*a*) $A + B$ (*b*) $\overline{A}.\overline{B}$ (*c*) B (*d*) A

39. Truth table for given circuit is:



(*a*)

A	*B*	*Y*
0	0	1
0	1	0
1	0	1
1	1	0

(*b*)

A	*B*	*Y*
0	0	1
0	1	0
1	0	0
1	1	1

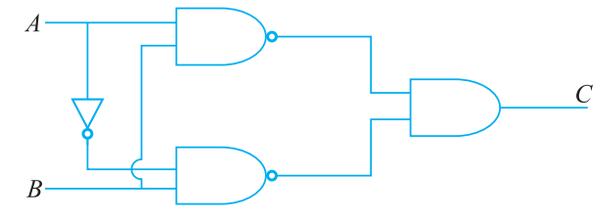
(*c*)

A	*B*	*Y*
0	0	0
0	1	0
1	0	0
1	1	0

(*d*)

A	*B*	*Y*
0	0	0
0	1	1
1	0	1
1	1	0

40.



The truth table for the given logic circuit is

(*a*)

A	*B*	*C*
0	0	1
0	1	0
1	0	0
1	1	1

(*b*)

A	*B*	*C*
0	0	1
0	1	0
1	0	1
1	1	0

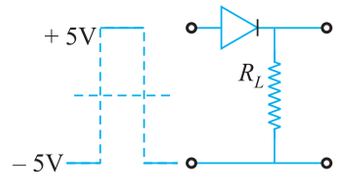
(*c*)

A	*B*	*C*
0	0	0
0	1	0
1	0	0
1	1	1

(*d*)

A	*B*	*C*
0	0	0
0	1	1
1	0	1
1	1	0

41. In a *p-n* junction, a square input signal of 10 V is applied, as shown



Then the output across R_L will be

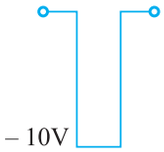
(*a*)



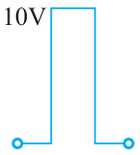
(*b*)



(*c*)



(*d*)



42. Given below are two statements: One is labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A): The OR gate function corresponds to switches in parallel.

Reason (R): The AND gate function corresponds to switches in series.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

Samiconductor, Metal and Non-Metal

43. The resistivity of a semiconductor depends upon

(*a*) Size of the atom (*b*) The nature of atoms

(*c*) Type of bonds (*d*) Size and types of motion

44. A pure semiconductor has

(*a*) An infinite resistance at 0°C

(*b*) A finite resistance which does not depend upon temperature

(*c*) A finite resistance which increases with temperature

(*d*) A finite resistance which decreases with temperature

45. If the ratio of the concentration of electrons to that of hole in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities?

(*a*) $\frac{5}{8}$ (*b*) $\frac{4}{5}$ (*c*) $\frac{5}{4}$ (*d*) $\frac{4}{7}$

46. Choose the correct option

(*a*) $\mu = v_d/E$, μ: mobility ($m^2/V\ s$); v_d: drift speed, E: Electric field

(*b*) Conductivity of conductors; $\sigma = n_e \mu_e e$

(*c*) Conductivity of semiconductors; $\sigma = n_e \mu_e e + n_h \mu_h e$; n_e: free electron density, n_h: hole density

(*d*) All options are correct

47. Given below are two statements: One is labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A): Temperature coefficient of resistance for semiconductor is negative.

Reason (R): Conductivity of semiconductor increases with increment in temperature

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

48. The semiconductors are generally

(*a*) Monovalent (*b*) Divalent

(*c*) Trivalent (*d*) Tetravalent

Conduction Band

49. The effect of increase in temperature on the number of electrons in conduction band (n_e) and resistance of a semiconductor will be as:

(*a*) Both n_e and resistance decrease

(*b*) Both n_e and resistance increase

(*c*) n_e increases, resistance decreases

(*d*) n_e decreases, resistance increases

50. In semiconductors at a room temperature

(*a*) The valence band is partially empty, and the conduction band is partially filled

(*b*) The valence band is completely filled, and the conduction band is partially filled

(*c*) The valence band is completely filled

(*d*) The conduction band is completely empty

51. In pure semiconductor at zero kelvin temperature

(*a*) The valence band is completely filled and the conduction band partially filled

(*b*) Valence band completely filled and conduction band completely empty

(*c*) Both valence band and conduction band empty

(*d*) The valence band is partially empty and the conduction band is partially filled

52. Choose the only false statement from the following

(*a*) In conductors the valence and conduction bands overlap

(*b*) Substance with energy gap of the order 10 eV are insulators

(*c*) The resistivity of a semiconductor increases with increase in temperature

(*d*) The conductivity of a semiconductor increases with increasing temperature

53. In semiconductors, which of the following relations is correct at thermal equilibrium?

(*a*) $n_i = n_e = n_h$

(*b*) $n^2 = n_e n_h$

(*c*) $n_i = n_e/n_h$

(*d*) $n_i = n_e + n_h$

54. In an intrinsic (pure) semiconductor, the number of conduction electrons is 7×10^{19} per cubic metre. Find the total number of current carriers (electrons and holes) in a same semiconductor of size 1 cm × 1 cm × 1 mm.

Doping

55. When *n*-type semiconductor is heated

(*a*) Number of electrons increases while that of holes decreases

(*b*) Number of holes increases while that of electrons decreases

(*c*) Number of electrons and holes remain same

(*d*) Number of electrons and holes increases equally

56. The impurity atoms with which pure silicon should be doped to make a *p*-type semiconductor are those of

(*a*) Phosphorus

(*b*) Antinomy

(*c*) Boron

(*d*) Copper

57. **Statement I:** By doping silicon semiconductor with pentavalent material, the electron density increases.

Statement II: The *n*-type semiconductor has net negative charge.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Statement I is false but Statement II is true

(*b*) Statement I is true but Statement II is false

(*c*) Both Statement I and Statement II are true

(*d*) Both Statement I and Statement II is false

58. Match the List-I with List-II

List-I		List-II	
A.	Intrinsic Semiconductor	I.	Fermi-level near conduction band
B.	n-type semiconductor	II.	Fermi-level at middle
C.	p-type semiconductor	III.	Fermi-level near valence band
D.	Metals	IV.	Fermi-level inside conduction band

(*a*) A → I, B → II, C → III, D → IV

(*b*) A → II, B → I, C → III, D → IV

(*c*) A → II, B → III, C → I, D → IV

(*d*) A → III, B → I, C → II, D → IV

59. Pure Si at 500K has equal number of electrons (n_e) and holes (n_h) concentrations of 1.5×10^{16} m^{-3}. Doping by Indium increases (n_h) to 4.5×10^{22} m^{-3}. The doped semiconductor is of

(*a*) *p*-type having electron concentration (n_e) = 5×10^{9} m^{-3}

(*b*) *n*-type with electron concentration (n_e) = 5×10^{22} m^{-3}

(*c*) *p*-type with electron concentration (n_e) = 2.5×10^{10} m^{-3}

(*d*) *n*-type with electron concentration (n_e) = 2.5×10^{23} m^{-3}

60. In an intrinsic semiconductor, the density of conduction electrons is 7.07×10^{15} m^{-3}. When it is doped with indium, the density of holes becomes 5×10^{22} m^{-3}. Find the density of conduction electrons in doped semiconductor:

(*a*) Zero (*b*) 1×10^{9} m^{-3}

(*c*) 7×10^{15} m^{-3} (*d*) 5×10^{22} m^{-3}

61. Pure Si at 300 K has equal electron (n_e) and hole (n_h) concentration of 2×10^{16} per m^3. Doping by indium increases n_h to 4×10^{22} per m^3. Calculate n_e in the doped silicon.

62. In a pure silicon ($n_1 = 10^{16}$/m^3) crystal at 300 K, 10^{21} atoms of phosphorus are added per cubic meter. The new hole concentration will be:

(*a*) 10^{21} per m^3 (*b*) 10^{19} per m^3

(*c*) 10^{11} per m^3 (*d*) 10^{5} per m^3

63. For a *p*-type semiconductor which of the following statement is true?

(*a*) Electrons are the majority carriers and pentavalent items are dopants

(*b*) Electrons are the majority carrier and trivalent atoms are the dopants

(*c*) Holes are the majority carriers and trivalent atoms are the dopants

(*d*) Holes are the majority carriers and pentavalent atoms are the dopants

64. In a *n*-type semiconductor which of the following statement is true?

(*a*) Holes are minority carriers and pentavalent atoms are dopants

(*b*) Holes are majority carriers and trivalent atoms are dopants

(*c*) Electrons are majority carriers and trivalent atoms are dopants

(*d*) Electrons are minority carriers and pentavalent atoms are dopants

65. Match the List-I with List-II

List-I (Type of semiconductor)		List-II (Properties)	
(*i*)	Intrinsic semiconductor	(1)	Insulator at room temperature
(*ii*)	*N*-type semiconductor	(2)	$n_h >> n_e$
(*iii*)	*P*-type semiconductor	(3)	$n_e >> n_h$
		(4)	$n_e = n_h = n_i$

(*a*) (*i*) → 1, (*ii*) → 2, (*iii*) → 3

(*b*) (*i*) → 2, (*ii*) → 3, (*iii*) → 4

(*c*) (*i*) → 3, (*ii*) → 4, (*iii*) → 1

(*d*) (*i*) → 4, (*ii*) → 3, (*iii*) → 2

66. Given below are two statements: One is labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A): The number of electrons in *n*-type silicon semiconductor is higher than the number of electrons in a pure silicon semiconductor.

Reason (R): The law of mass action is applicable only to *n*-type semiconductors.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

p-n Junction

67. **Assertion (A):** Diffusion current in a *p-n* junction is greater than the drift current in magnitude if the junction is forward biased.

Reason (R): Diffusion current in a *p-n* junction is from the *n*-side to the *p*-side if the junction is forward biased.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Assertion (A) is false but Reason (R) is true.

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Both Assertion (A) and Reason (R) are true but Reason (R) is the correct explanation of Assertion (A).

68. The barrier potential of a *p-n* junction depends on

(1) Type of semiconductor material

(2) Amount of doping

(3) Temperature

Which one of the following is correct?

(*a*) (1) and (2) only (*b*) (2) only

(*c*) (2) and (3) only (*d*) (1), (2) and (3)

69. In a *p-n* junction

(*a*) High potential at *n* side and low potential at *p* side

(*b*) High potential at *p* side and low potential at *n* side

(*c*) *p* and *n* both are at same potential

(*d*) Undetermined

70. Direction of diffusion current in *P-N* junction diode is:

(*a*) From P side to N-side (*b*) From N side to P-side

(*c*) Randomly oriented (*d*) From both side one by one

71. Depletion layer consists of

(*a*) mobile ions (*b*) protons

(*c*) electrons (*d*) immobile ions

72. In a *p-n* junction diode, change in temperature due to heating

(*a*) affects only reverse resistance

(*b*) affects only forward resistance

(*c*) does not affect resistance of *p-n* junction

(*d*) affects the overall *V-I* characteristics of *p-n* junction

73. In a *p-n* junction photo cell, the value of the photo-electromotive force produced by monochromatic light is proportional to

(*a*) The barrier voltage at the *p-n* junction

(*b*) The intensity of the light falling on the cell

(*c*) The frequency of the light falling on the cell

(*d*) The voltage applied at the *p-n* junction

74. Which of the following statements is not true?

(*a*) The resistance of intrinsic semiconductors decrease with increase of temperature

(*b*) Doping pure Si with trivalent impurities gives *p*-type semiconductors

(*c*) The majority carriers in *N*-type semiconductors are holes

(*d*) A *PN*-junction can act as a semiconductor diode

75. Given below are two statements: One is labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A): In an unbiased *p-n* junction, holes diffuse from the *p*-region to *n*-region.

Reason (R): Hole concentration in *p*-region is more as compared to *n*-region.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

76. Which of the given statements are correct regarding to formation of unbiased *P-N* junction diode.

(I) Drift and diffusion currents occur *P* to *N* side.

(II) Initially diffusion current is large and drift current is very small.

(III) Finally diffusion and drift currents grow to equal in magnitude.

(IV) Under equilibrium there is no net current across *P-N* junction.

(*a*) I and IV (*b*) I, II and III

(*c*) II, III and IV (*d*) All of these

77. **Statement I:** In equilibrium condition. *P* side of a *P-N* junction is at positive potential.

Statement II: A *P*-type semiconductor contains more holes than free electrons.

(*a*) Both Statement I and Statement II are correct.

(*b*) Statement I is correct and Statement II is incorrect.

(*c*) Statement I in incorrect and Statement II is correct.

(*d*) Both Statement I and Statement II are incorrect.

Forward and Reverse Biased

78. The increase in the width of the depletion region in a *p-n* junction diode is due to

(*a*) Forward bias only

(*b*) Reverse bias only

(*c*) Both forward bias and reverse bias

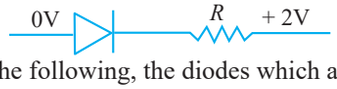
(*d*) Increase in forward current

79. Which one of the following represents forward bias diode?

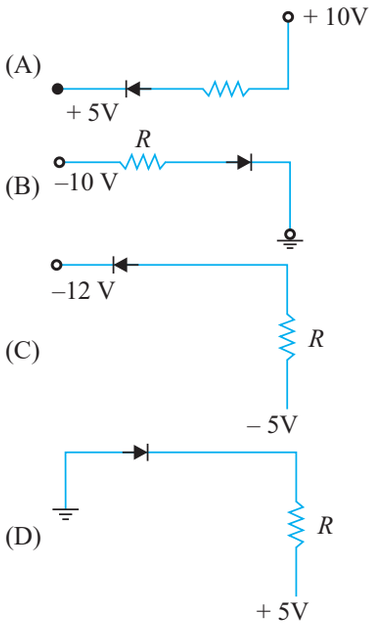
(*a*) 

(*b*) 

(*c*) 

(*d*) 

80. In the following, the diodes which are forward biased are



(*a*) (A), (B) and (D) (*b*) (C) only

(*c*) (C) and (A) (*d*) (B) and (D)

81. In forward biasing of the *p*-*n* junction

(*a*) The positive terminal of the battery is connected to *p*-side and the depletion region becomes thick

(*b*) The positive terminal of the battery is connected to *n*-side and the depletion region becomes thin

(*c*) The positive terminal of the battery is connected to *n*-side and the depletion region becomes thick

(*d*) The positive terminal of the battery is connected to *p*-side and the depletion region becomes thin

82. Select incorrect alternative

(*a*) Diffusion current is due to motion of the majority charge carriers.

(*b*) Drift current is due to motion of the minority charge carriers.

(*c*) If diffusion current is less than drift current the *p*-*n* junction is forward biased.

(*d*) All of the above are correct.

83. Reverse bias applied to a junction diode

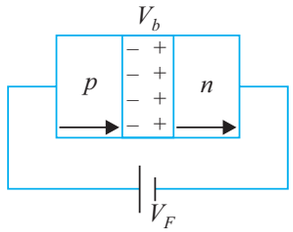
(*a*) Lowers the potential barrier

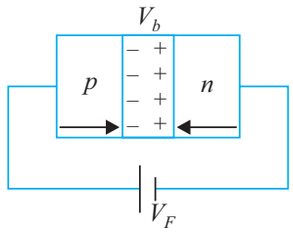
(*b*) Raises the potential barrier.

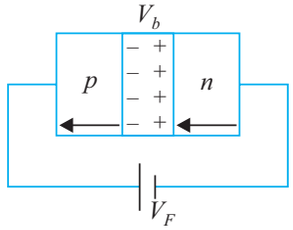
(*c*) Increases the majority carrier current

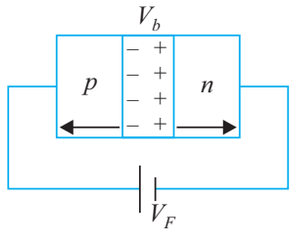
(*d*) Increases the minority carrier current

84. In the case of forward biasing of *p*-*n* junction, which one of the following figures correctly depicts the direction of flow of carriers?

(*a*) 

(*b*) 

(*c*) 

(*d*) 

85. **Assertion (A):** Width of depletion region is reduced in forward bias.

Reason (R): In *N* type semiconductor majority charge carriers are free electrons while in *P* type they are holes.

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

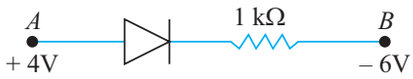
(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

Diode

86. Consider the junction diode as ideal. The value of current flowing through AB is



(*a*) 10^{-1} *A* (*b*) 10^{-3} *A*

(*c*) 0 *A* (*d*) 10^{-2} *A*

87. For using a multimeter to identify diode from electrical components. Choose the correct statement out of the following about the diode

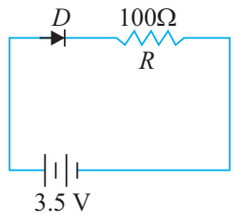
(*a*) It is two terminal device which conducts current in both directions

(*b*) It is two terminal device which conducts current in one direction only

(*c*) It does not conduct current gives an initial deflection which decays to zero

(*d*) It is three terminal device which conducts current in one direction only between central terminal and either of the remaining two terminals

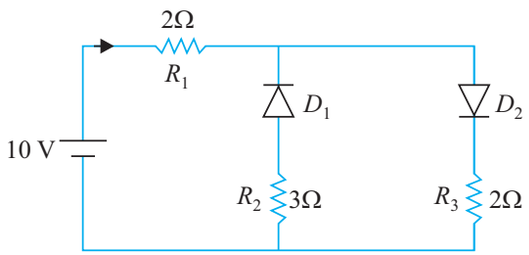
88. In the given figure, a diode D is connected to an external resistance $R = 100\ \Omega$ and an e.m.f. of 3.5 *V*. If the barrier potential developed across the diode is 0.5 *V*, the current in the circuit will be



(*a*) 20 mA (*b*) 35 mA

(*c*) 30 mA (*d*) 40 mA

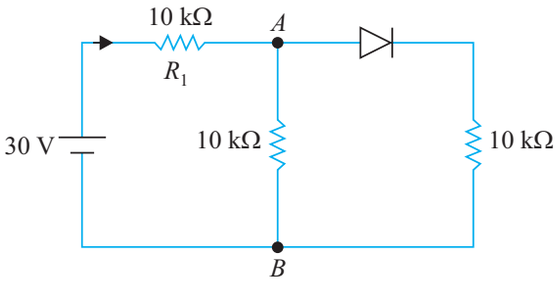
89. The given circuit has two ideal diodes connected as shown in figure. The current flowing through the resistance R_1 will be



(*a*) 2.5 A (*b*) 10.0 A

(*c*) 1.43 A (*d*) 3.13 A

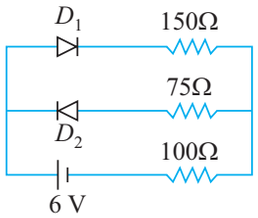
90. In the figure, potential difference between A and B is



(*a*) 5 V (*b*) 10 V

(*c*) zero (*d*) 15 V

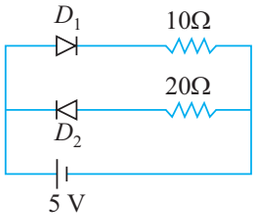
91. The circuit shown below contains two ideal diodes, each with a forward resistance of 50 Ω. If the battery voltage is 6 V, the current through the 100 Ω resistance (in amperes) is



(*a*) 0.036 (*b*) 0.020

(*c*) 0.027 (*d*) 0.030

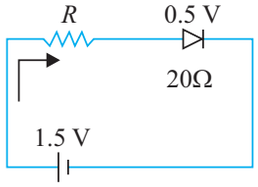
92. Two ideal diodes are connected to a battery as shown in the circuit. The current supplied by the battery is



(*a*) 0.75 A (*b*) zero

(*c*) 0.25 A (*d*) 0.5 A

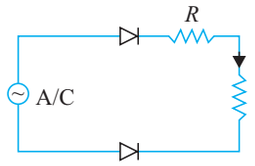
93. The diode used in the circuit shown in the figure has a constant voltage drop at 0.5 V at all currents and a maximum power rating of 100 mili watts. What should be the value of the resistor R, connected in series with diode for obtaining maximum current?



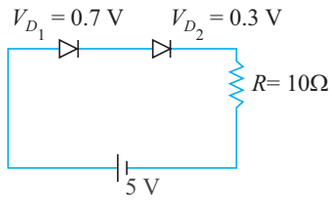
(*a*) 6.76 Ω (*b*) 20 Ω

(*c*) 40 Ω (*d*) 5.6 Ω

94. Find current flowing through resistance?

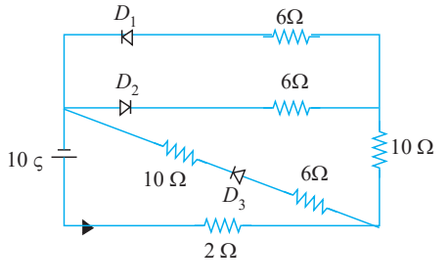


95. Find current flowing through resistor

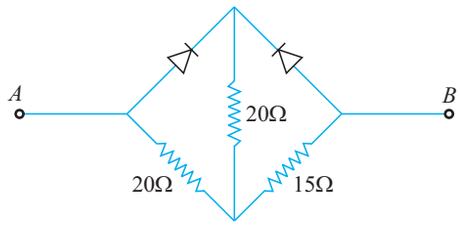


(*a*) $I = \frac{5}{10} A$ (*b*) $I = \frac{4}{10} A$ (*c*) $I = 0$ (*d*) $I = \frac{6}{10} A$

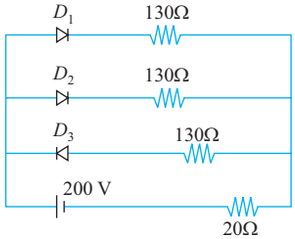
96. As per the given circuit, the value of current through the battery will be _____ A.



97. Two ideal diodes are connected in the network as shown is figure. The equivalent resistance between A and B is _____ Ω.

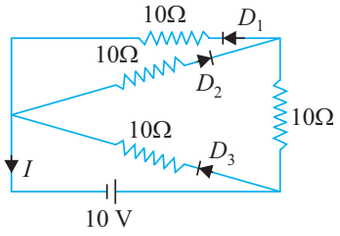


98. In the given figure, each diode has a forward bias resistance of 30 Ω and infinite resistance in reverse bias. The current I_1 will be.



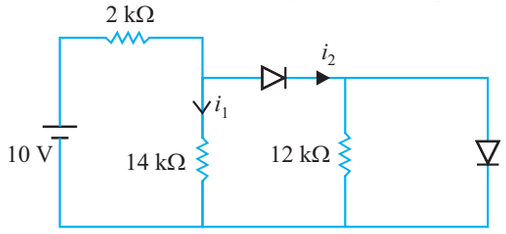
(*a*) 3.75 A (*b*) 2.76 A (*c*) 2.35 A (*d*) 2 A

99. In the given circuit, the current (*I*) through the battery will be:



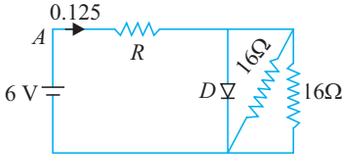
(*a*) 1.5 A (*b*) 1 A (*c*) 2.5 A (*d*) 2 A

100. In the following circuit, find i_1 and current i_2



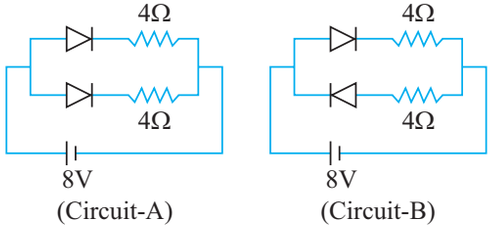
(*a*) 0, 0 (*b*) 5 mA, 5 mA
(*c*) 5 mA, 0 (*d*) 0 & 5mA

101. In the given circuit, the diode has a forward resistance of 8 Ω and infinite backward resistance. The value of *R* is:



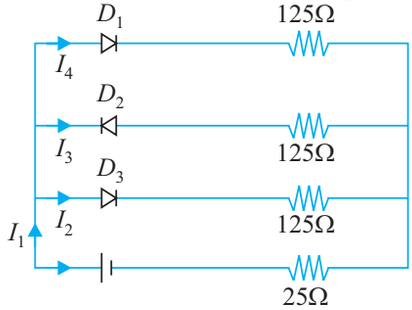
(*a*) 44 Ω (*b*) 48 Ω (*c*) 16 Ω (*d*) 8 Ω

102. Current flowing in each of the circuit A and B respectively are:



(*a*) 1 Amp, 2 Amp (*b*) 1 Amp, 4 Amp
(*c*) 4 Amp, 2 Amp (*d*) 2 Amp, 4 Amp

103. If each diode has a forward bias resistance of 25Ω in the below circuit. Which of the following options is correct:



(*a*) $\frac{I_3}{I_4} = 1$ (*b*) $\frac{I_2}{I_4} = 1$ (*c*) $\frac{I_1}{I_2} = 1$ (*d*) $\frac{I_1}{I_3} = 2$

Photodiode

104. **Assertion (A):** Photodiodes are preferably operated in reverse bias condition for light intensity measurement.

Reason (R): The current in the forward bias in more than the current in the reverse bias for a *p-n* junction diode.

In the light of the above statement, choose the correct answer from the options given below:

(*a*) Assertion (A) is false but Reason (R) is true.

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A)

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Both Assertion (A) and Reason (R) are true but Reason (R) is the correct explanation of Assertion (A).

105. The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because

(*a*) Fractional change in majority carriers produce higher forward bias current.

(*b*) Fraction change in majority carriers produce higher reverse bias current.

(*c*) Fractional change in majority carriers produce higher forward bias current.

(*d*) Fractional change in minority carriers produce higher reverse bias current.

106. Symbolic representation of photodiode is

(*a*) 

(*b*) 

(*c*) 

(*d*) 

107. **Assertion (A):** A *P-N* Photodiode is made from a semiconductor for which $E_g = 2.8$ eV. This photo diode will not detect the wavelength of 6000 nm.

Reason (R): A *PN* Photodiode detect wavelength λ if $\frac{hc}{\lambda} > E_g$

(*a*) Assertion (A) is false but Reason (R) is true.

(*b*) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Both Assertion (A) and Reason (R) are true but Reason (R) is the correct explanation of Assertion (A).

Zener Diode

108. Choose the correct statement about Zener diode:

(*a*) It works as a voltage regulator in reverse bias and behaves like simple *p-n* junction diode in forward bias

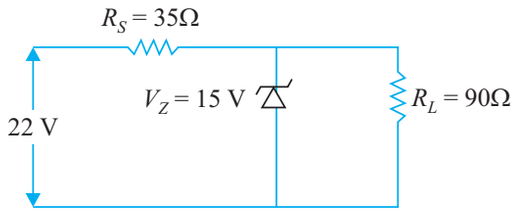
(*b*) It works as a voltage regulator in both forward and reverse bias

(*c*) It works as a voltage regulator only in forward bias

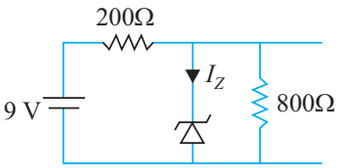
(*d*) It works as a voltage regulator in forward bias and behaves like simple *p-n* junction diode in reverse bias

109. The value of power dissipated across the zener diode (V_z = 15 V) connected in the circuit as shown in the figure is $x \times 10^{-1}$ watt.

The value of x to the nearest integer is _____.



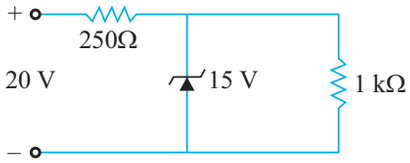
110. The reverse breakdown voltage of a Zener diode is 5.6 V in the given circuit.



(*a*) 10 mA (*b*) 15 mA

(*c*) 7 mA (*d*) 17 mA

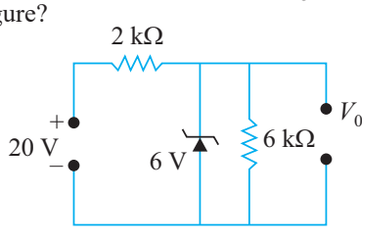
111. A Zener diode, having breakdown voltage equal to 15 V, is used in a voltage regulator circuit shown in figure. The current through the diode is



(*a*) 5 mA (*b*) 10 mA

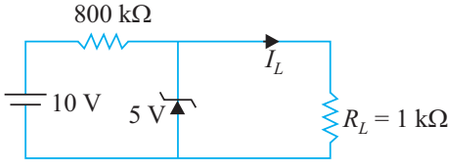
(*c*) 15 mA (*d*) 20 mA

112. What is the value of output voltage V_0 in the circuit shown in the figure?



(*a*) 6 V (*b*) 14 V (*c*) 20 V (*d*) 26 V

113. In the given circuit, the value of current I_L will be _____ mA. (When R_L = 1 kΩ)



114. **Assertion (A):** Zener diode works on a principle of breakdown voltage.

Reason (R): Current increases suddenly after breakdown voltage.

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is NOT the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

LED

115. Which of the following statement is not correct in the case of light emitting diodes?

(A) It is a heavily doped *p-n* junction

(B) It emits light only when it is forward biased.

(C) It emits light only when it is reverse biased.

(D) The energy of the light emitted is equal to or slightly less than the energy gap of the semiconductor used

Choose the correct answer from the options given below:

(*a*) (C) and (D) (*b*) (A) only

(*c*) (C) only (*d*) (B) only

116. An LED is constructed from a *p-n* junction diode using GaAsP. The energy gap is 1.9 eV. The wavelength of the light emitted will be equal to

(*a*) 10.4×10^{-26} m (*b*) 654 nm

(*c*) 654 Å (*d*) 654×10^{-11} m

117. A light emitting diode (LED) has a voltage drop of 2 V across it and passes a current of 10 mA, when it is operated with a 6 V battery through a limiting resistor *R*. The value of *R* is:

(*a*) 100 Ω (*b*) 200 Ω (*c*) 400 Ω (*d*) 600 Ω

118.

Column-I		Column-II	
(A)	LED or diode lasers	(1)	Detectors (Demodulator) in optical communication
(B)	Semiconductor photodiode	(2)	Optical source in optical communication
(C)	Transducer	(3)	Conversion of light signal into electrical signal
(D)	Photodetectors	(4)	Conversion of information (Physical quality into electrical signal)

Codes:

	(A)	(B)	(C)	(D)
(*a*)	2	1	4	3
(*b*)	4	3	2	1
(*c*)	3	2	5	4
(*d*)	2	4	1	3

119. Match the column-I of special dioes with their utilization given in column-II:

Column-I		Column-II	
(*a*)	Zener diode	(1)	As a light sensor
(*b*)	LED	(2)	Voltage regulator circuit
(*c*)	Photodiode	(3)	As a bulb of light

(*a*) (*a*) → 3, (*b*) → 2, (*c*) → 1

(*b*) (*a*) → 3, (*b*) → 1, (*c*) → 2

(*c*) (*a*) → 1, (*b*) → 3, (*c*) →

(*d*) (*a*) → 2, (*b*) → 3, (*c*) → 1

120. In case of LED if we increase forward applied voltage:

(*a*) Intensity of emitted light increases

(*b*) Intensity of emitted light decreases

(*c*) Intensity of emitted light first increases then decreases

(*d*) Intensity of emitted light first decreases then increases

Full wave Rectifier & Half wave Rectifier

121. **Statement I:** To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load R_L.

Statement II: To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with R_L

In the light of the above statements, choose the most appropriate answer from the options given below:

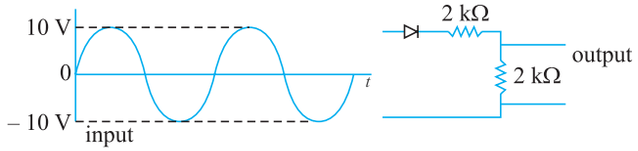
(*a*) Statement I is false but Statement II is true

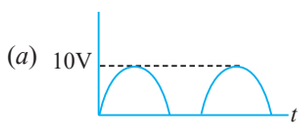
(*b*) Statement I is true but Statement II is false

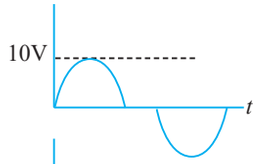
(*c*) Both Statement I and Statement II are true

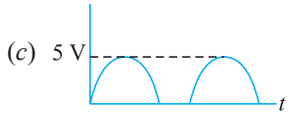
(*d*) Both Statement I and Statement II is false

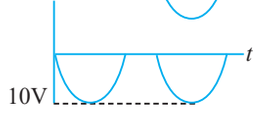
122. Find out wave from:



(*a*) 

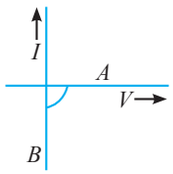
(*b*) 

(*c*) 

(*d*) 

V-I Characteristic Graph

123. The given graph represents V–I characteristics for a semiconductor device. Which of the following statement is correct?



(*a*) It is V–I characteristics for a solar cell, where, point *A* represents open circuit voltage and point *B* short circuit current.

(*b*) It is for a solar cell and points *A* and *B* represent open circuit voltage and current, respectively.

(*c*) It is for a photodiode and points *A* and *B* represent open circuit voltage and current respectively.

(*d*) It is for a LED and points *A* and *B* represent open circuit voltage and short circuit current respectively.

MR* CORNER

1. **Statement I:** Electric conduction in a semiconductor takes place due to both electrons & holes.

Statement II: An electric field is applied to a Semiconductor. The number of charge carriers *n* will increase & the average drift speed *v* will decrease if the Temperature is increased.

2. **Statement I:** n_P (number of holes) and n_e (number of conduction electrons) are equal in an intrinsic semiconductor.

Statement II: n_P is lesser than n_e in any type of an extrinsic semiconductor.

3. **Statement I:** A *p*-Type semiconductor is uncharged at 0K but charged at higher temperatures.

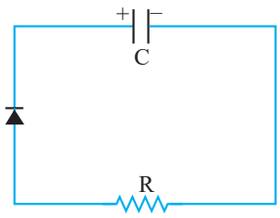
Statement II: If The two ends of a *p-n* Junction are joined by a wire, then there will not be a steady current in the circuit

4. **Statement I:** The drift current in a *p-n* junction is from *n*-side to *p*-side if junction is forward biased and in opposite direction if it is reverse biased

Statement II: The diffusion current in a *p-n* junction is from the *p*-side to the *n*-side

5. **Statement I:** Diffusion current in a *p-n* Junction is greater than the drift current in magnitude only if the junction is forward-biased.

Statement II: The capacitor of capacitance C is charged to the potential *V* and is connected in the following circuit as shown at $t = 0$. The charge on the capacitor at time $t = RC$ will be CV/e.



6. **Statement I:** A hole diffuses from the *p*-side to *n*-side in *p-n* junction. This means that a bond is broken on *n*-side & e^- freed from the bond jumps to broken bond on the *p*-side to complete it

Statement II: In a *p-n* junction, new holes & conduction electrons are produced continuously throughout the material but holes & conduction electrons recombine continuously throughout material except in the depletion region.

7. **Statement I:** In a semiconductor, there are no free electrons at 0°C, but number of free electrons increases with temperature.

Statement II: The electrical conductivity of pure germanium can only be increased by doping acceptor impurities

8. **Statement I:** A semiconductor is doped with a donor impurity. The hole concentration will decrease while electron concentration increases.

Statement II: An AND gate can be prepared by repetitive use of NAND gate.

Answer Key

1. (*a*)	**2.** (*c*)	**3.** (*a*)	**4.** (*d*)	**5.** (*b*)	**6.** (*b*)	**7.** (*a*)	**8.** (*a*)	**9.** (*a*)	**10.** (*a*)
11. (*b*)	**12.** (*a*)	**13.** (*a*)	**14.** (*b*)	**15.** (*a*)	**16.** (*b*)	**17.** (*c*)	**18.** (*a*)	**19.** (*a*)	**20.** (*d*)
21. (*c*)	**22.** (*b*)	**23.** (*a*)	**24.** AND gate		**25.** (*b*)	**26.** X–NOR gate		**27.** (*d*)	**28.** NOR
29. (*b*)	**30.** (*b*)	**31.** (*c*)	**32.** (*c*)	**33.** (*b*)	**34.** (*d*)	**35.** (*d*)	**36.** (*a*)	**37.** (*d*)	**38.** (*c*)
39. (*c*)	**40.** (*b*)	**41.** (*b*)	**42.** (*b*)	**43.** (*b*)	**44.** (*d*)	**45.** (*c*)	**46.** (*d*)	**47.** (*a*)	**48.** (*d*)
49. (*c*)	**50.** (*a*)	**51.** (*b*)	**52.** (*c*)	**53.** (*b*)	**54.** 14×10^{12}		**55.** (*d*)	**56.** (*c*)	**57.** (*b*)
58. (*b*)	**59.** (*a*)	**60.** (*b*)	**61.** 10^{10} per m^3		**62.** (*c*)	**63.** (*c*)	**64.** (*a*)	**65.** (*d*)	**66.** (*c*)
67. (*c*)	**68.** (*d*)	**69.** (*a*)	**70.** (*a*)	**71.** (*d*)	**72.** (*d*)	**73.** (*b*)	**74.** (*c*)	**75.** (*a*)	**76.** (*c*)
77. (*c*)	**78.** (*b*)	**79.** (*d*)	**80.** (*c*)	**81.** (*d*)	**82.** (*d*)	**83.** (*b*)	**84.** (*b*)	**85.** (*b*)	**86.** (*d*)
87. (*b*)	**88.** (*c*)	**89.** (*a*)	**90.** (*b*)	**91.** (*b*)	**92.** (*d*)	**93.** (*c*)	**94.** I = 0	**95.** (*b*)	**96.** 1 A.
97. 25 Ω.	**98.** (*d*)	**99.** (*a*)	**100.** (*d*)	**101.** (*a*)	**102.** (*c*)	**103.** (*c*)	**104.** (*b*)	**105.** (*d*)	**106.** (*c*)
107. (*d*)	**108.** (*a*)	**109.** (*x* = 5)	**110.** (*a*)	**111.** (*a*)	**112.** (*a*)	**113.** 5 mA	**114.** (*b*)	**115.** (*c*)	**116.** (*b*)
117. (*c*)	**118.** (*a*)	**119.** (*d*)	**120.** (*c*)	**121.** (*b*)	**122.** (*c*)	**123.** (*a*)			

MR* CORNER

1. Both True	**2.** I. True, II. Flase	**3.** I. Flase, II. True	**4.** I. Flase, II. True	**5.** I. True, II. False
6. Both True	**7.** Both Flase	**8.** Both True.		